Safety-Driven and Localization Uncertainty-Driven Perception-Aware Trajectory Planning for Quadrotor Unmanned Aerial Vehicles

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***Abstract*— Recent advances in trajectory planning have enabled quadrotor unmanned aerial vehicles (UAVs) to navigate autonomously in complex environments. However, most of the existing methods do not consider the perception quality and the safety simultaneously. This article proposes a perception-aware trajectory planning strategy for quadrotors, which can ensure the safety and localization accuracy. In contrast to the existing methods, the main idea of the proposed method lies in that the yaw angle trajectory is planned to actively obtain more information in the environment to improve the localization accuracy and keep the safe flight simultaneously. Following the mainstream two-stage motion planning framework, a coarse-to-fine graph search strategy is proposed to search for a safe and perception-aware yaw angle path in the first stage. Specifically, a *Yaw Safety Corridor* (YSC) is proposed to guarantee the safety, which can observe the obstacles directly along the tangent direction of the position trajectory. In addition, a dedicated map *Fisher Information Field* (FIF) is employed to evaluate the perception quality. In the second stage, a path-guided optimization method is proposed to quickly generate a safe and perception-aware trajectory. Finally, comparative simulation and real-world experiments are conducted to verify the superior performance in terms of the perception quality and the safety of the proposed method.**

***Index Terms*— Perception awareness, YSC, localization uncertainty, coarse-to-fine search.**

I. Introduction

UNMANNED aerial vehicles (UAVs), especially quadrotors, equipped with the visual sensors, are suitable for different tasks such as mapping [1], transportation [2], [3], autonomous exploration [4], [5], [6], [7], inspection [8] and rescue [9]. However, it is essential to design robust localization [10] and planning algorithms to keep UAVs safely navigating in complex environments. Visual-inertial state estimation has received much attention in recent years [11], which assists the vision system with a low-cost inertial measurement (IMU) [12] for the localization. Typical visual-inertial navigation systems such as SVO [13], OKVIS [14], VINS [15] and ORB-SLAM3 [16] all track multiple visual features in the keyframes and adopt nonlinear optimization technique to estimate the state of the robot. The quantity and the quality of the visual features in the view will affect the localization quality, resulting in the localization uncertainty. For the planning algorithm, most of the existing methods focus on the safety of the position trajectory and time-optimal performance [17], [18], [19], [20]. However, the localization uncertainty is not considered as shown in Fig. 1. During the flight, the motion planning algorithm, without considering the localization uncertainty, potentially results in facing featureless areas such as white walls, which may cause a localization failure. Thus, a catastrophic crash may happen. However, the trajectory considering the localization uncertainty can keep more visual landmarks in the view to increase the localization accuracy as shown in Fig. 2. Therefore, it is necessary to consider the perception quality in the motion planning, which is called *perception-aware planning*, to improve the performance on the localization accuracy.

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A. Motivation

Although some methods have made certain progress in perception-aware planning, there are still some problems in the actual applications, which are listed as follows:

1) Most of the existing methods either aim to guarantee the safety without considering the perception awareness, or consider the perception awareness without guaranteeing the safety. In addition, the safety of the yaw angle trajectory is ignored or formulated as a soft constraint, which can not strictly bound the yaw angle in a safe region.
2) The mainstream framework decomposes the motion planning into a graph search process and a trajectory

optimization process. To get a fine yaw path, many nodes are added to a directed graph, which is computationally intensive. At the same time, current methods generally optimize a trajectory by a gradient-descent method, which can be easily trapped in local minima.

B. Contributions

To address the aforementioned issues, we propose a **S**afety-driven and **L**ocalization **U**ncertainty-driven **P**erception-aware trajectory generation method (SLUP). At first, a yaw safety corridor is constructed according to the position trajectory to evaluate the safety of the yaw angle. Next, a coarse-to-fine graph search approach, which efficiently avoids abundant useless search, is proposed to first search a relatively rough path in the yaw safety corridor. Then, a fine path is obtained in the vicinity of the rough path, which is strictly bound in the yaw safety corridor, and the localization uncertainty is considered simultaneously. The trajectory is parameterized to the uniform B-spline, and the convex hull property of the B-spline is utilized to satisfy the safety constraint. Finally, a path-guided optimization method is proposed to generate a yaw angle trajectory, which can eliminate the local minima problem and balance the safety and perception quality effectively. The contributions of this article are summarized as follows:

1) A universal yaw safety metric is put forward to evaluate the safety of the yaw angle during the planning, which is employed to construct the yaw safety corridor for planning a safe trajectory.
2) A coarse-to-fine graph search strategy is proposed to avoid the abundant useless search, which efficiently searches a safe and perception-aware yaw angle path. After that, a path-guided optimization method is proposed to generate a safe and perception-aware trajectory, which fully utilizes the convex hull property of the B-spline and eliminates the local minima problem.
3) Comparative simulation and real-world tests are carried out to verify the superior performance in terms of the perception quality and the safety of the proposed method.

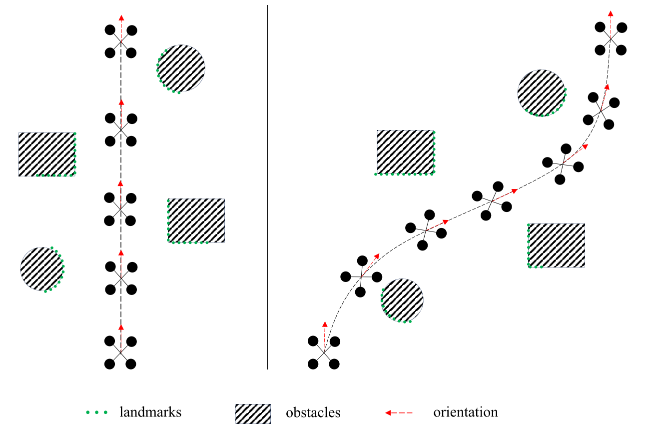


Fig. 1. Without considering the perception awareness, the planned trajectory is oriented along the tangent direction of the position trajectory. The texture information in the environment is not utilized, resulting in relatively considerable localization uncertainty.

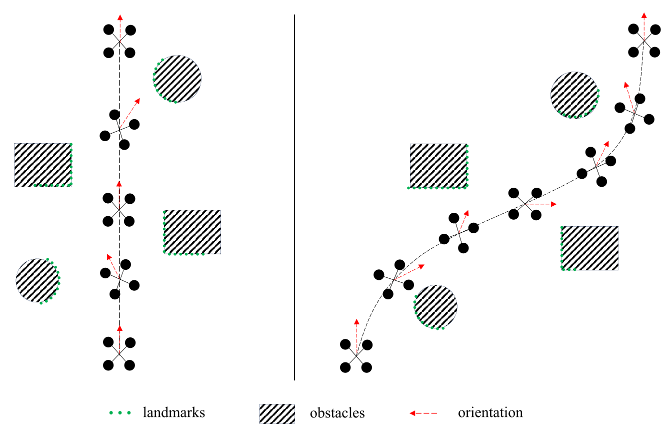


Fig. 2. Considering the perception awareness, the planned trajectory is oriented along the textured direction. The texture information in the environment is fully utilized to reduce the localization uncertainty.

II. Related Work

There are abundant methods on trajectory generation for quadrotors, which can be categorized into the hard-constrained methods and the soft-constrained methods. The hard-constrained methods construct equality or inequality constraints and solve a set of solutions that completely meet these constraints. The soft-constrained methods construct the penalty functions and minimize the value of the penalty functions through the optimization method to get a set of solutions. For the hard-constrained methods, in [21], trajectories are represented as piecewise polynomials, and the quadratic programming problem is formulated to generate a minimum snap trajectory. In [22], the minimum snap trajectory is solved in closed form, and waypoints are iteratively added to avoid being intersection with obstacles. Chen et al. [23] propose a corridor-based planner that plans safe trajectories in a series of connected cubes. After that, Liu et al. [24] utilize connected convex polygons to represent the corridor. To reduce the corridor generation time, Gao et al. [25] further represent the corridor with axis-aligned cubes. Apart from that, the corridor representation is generalized to convex free polyhedrons at a given coordinate [26]. Global optimality can be guaranteed in the hard-constrained methods. However they often can not meet the real-time performance due to the demand of iteratively adding more constraints to get a feasible solution.

Some methods formulate the trajectory generation problem as a gradient-based nonlinear optimization problem, which considers the safety and the smoothness as soft constraints. In the pioneering work, Ratliff et al. [27] utilize the gradient-descent method to optimize a discrete-time trajectory but suffers from the local minima problem. Oleynikova et al. [28] adopt the random restart method that relieves the local minima problem. Usenko et al. [29] use uniform B-spline to parameterize the trajectories to replan in real-time. However, the success rate is much low, and the trajectory is not completely safe. Zhou et al. [30] propose a GTO method that further exploits the properties of the B-spline to replan

TABLE I
REFERENCE SUMMARY

Formulation	Goal	Ref.
NPA	Minimize snap	[21]
	Avoid obstacles	[22]-[31]
PA	Keep visual features in the view	[32]-[34]
	Actively observe unknown environment	[35]-[38]
	Reduce localization uncertainty	[39]-[40], SLUP

more efficiently. However, since the cost is non-convex, the trajectory often gets stuck in infeasible local minima and causes a replanning failure. Traditionally, most of the exsiting optimization-based methods pre-build a Euclidean Signed Distance Functions (ESDF). In [31], an ESDF-free gradient-based local planning framework is proposed, which only computes the gradient when necessary. Soft-constrained methods fully utilize the gradient information while often suffer from the local minima problem. Above all, the common problem of these methods is that the perception quality is not considered in the planning.

As listed in Tab. I, we categorize the related work into perception-aware methods (PA) and not perception-aware methods (NPA). Moreover, the methods can be specifically classified according to the planning goal. Integrating the perception quality into planning has been gradually concerned in recent years. Watterson et al. [32] propose an on-manifold trajectory optimization method which jointly optimizes the position and the yaw angle to get a less conservative trajectory. However, it can not meet the real-time requirements. Falanga et al. [33] propose a real-time Model Predictive Control (MPC) that concentrates on keeping the visual features in the center of the image. Murali et al. [34] separately track position and yaw trajectory, which maximizes the co-visibility of the features to keep the features in the camera view from one keyframe to another, increasing the estimation accuracy. However, the above methods lack the obstacle avoidance. Lu et al. [35] propose a real-time perception-limited planning MPC method, which simultaneously plans obstacle-free trajectories and generates the trajectories with effective perception of the point of interest. Li et al. [36] propose a perception-constrained MPC method to guarantee the visibility of the payload during motion. Zhou et al. [37] propose a robust and perception-aware trajectory replanning system, which considers the safety, to actively observe the unknown environment. Wang et al. [38] propose a visibility-guided perception-aware trajectory framework for aerial tracking, which actively adjusts the relative distance between the quadrotor and the target to reduce the probability of the occlusion. However, the localization uncertainty is not considered. In [39], a perception-aware receding horizon system is proposed, which generates a series of candidate trajectories and gets the best one according to the perception quality, the position collision probability and the distance to the goal. Zhang and Scaramuzza [40] summarize the perception information into a discrete grid named Fisher Information Field (FIF). A perception-aware trajectory incorporating FIF is generated to improve the localization accuracy. However, the yaw angle safety is not considered. Motivated by the observations, this article proposes a safe and perception-aware trajectory generation method, which includes a coarse-to-fine path search strategy and a path-guided trajectory generation strategy.

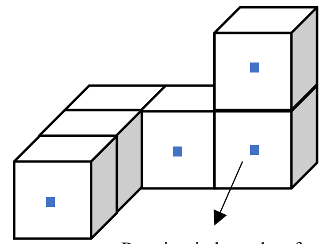


Fig. 3. Illustration of Fisher Information Field. Each black cube represents a voxel grid, which stores the rotation-independent component (blue squares). Fisher information at a certain pose can be computed in the constant time by a linear transformation.

III. PRELIMINARIES

A. Fisher Information Field

Fisher information is a crucial concept in the parameter estimation problem, which summarizes the information of the parameters to be estimated. A dedicated scene representation named FIF is employed to query the fisher information of a certain 6 degree-of-freedom (DoF) pose. Similar to ESDF for obstacle avoidance, FIF is also represented as the voxel grid. As shown in Fig. 3, for each voxel, a rotation-independent component of fisher information is stored, which is calculated from the visual features provided by the visual-inertial systems. Not all the visual features are chosen to calculate a rotation-independent component of fisher information. To construct the Fisher Information Field, we reserve a set of points from SVO output, which can be categorized into two groups: landmarks and seeds. Seeds are points whose position is still not estimated accurately. On the contrary, landmarks are points that have been observed multiple times from different frames and their positions have already been estimated accurately. Therefore, landmarks are chosen to build the Fisher Information Field. The field is updated in a batch offline. The process of constructing the field is rather intricate, as it requires both the scene depth information and the average view direction for each landmark. Besides, it needs to make a decision on whether a landmark can be matched from a given pose. It is challenging to incrementally build the field in real-time. Therefore, we generate a dedicated map offline, which subsequently is employed for efficient online planning. During the planning, the complete fisher information for 6 DoF pose can be recovered by the linear transformation of the rotation invariant component in constant time. The derivation details can be found in [40].

B. B-Spline

The B-spline curve [41] is adopted in the proposed trajectory optimization framework. Given $n+1$ control points $\boldsymbol{C} = \{C_0, C_1, \dots C_n\}$ and the knot vector $\boldsymbol{t} = [t_0, t_1, \dots, t_m]^{\mathbb{T}}$, in which $\boldsymbol{C}_i \in \mathbb{R}$ and $\boldsymbol{t}_i \in \mathbb{R}$. The B-spline curve function $p(t)$ is defined as follows:

$$p(t) = \sum_{i=0}^{n} C_i N_{i,d}, \tag{1}$$

where d is the degree of the curve. $N_{i,d}(t)$ is the B-spline blending function of the degree d, which can be computed recursively as follows:

$$N_{i,0} = \begin{cases} 1, & t_i \leq t \leq t_{i+1}, \\ 0, & else, \end{cases} \tag{2}$$

$$N_{i,d} = \frac{t - t_i}{t_{i+d} - t_i} N_{i,d-1} + \frac{t_{i+d+1} - t}{t_{i+d+1} - t_{i+1}} N_{i+1,d-1}, \tag{3}$$

An essential property of the B-spline is the convex hull property. The control points strictly bound the B-spline trajectory inside the convex hull. For the safety of the B-spline, all its convex hulls need to be collision-free. Furthermore, since the derivative of the B-spline is also the B-spline, the dynamic feasibility constraint, including the yaw angle velocity and the yaw angle acceleration, can be bound according to the convex hull property. The uniform B-spline is adopted in the proposed method, whose knot vector is uniformly distributed. In [30], further details are described.

IV. System Overview

The overview of the system is shown in Fig. 4, consisting of three modules: perception module, planning module and control module. The perception module utilizes the point clouds information provided by the depth camera to build ESDF of the surrounding environment for obstacle avoidance and FIF for evaluating the perception quality. The planning module generates a safe position trajectory and a safe, perception-aware, smooth and dynamically feasible yaw angle trajectory. The control module controls the quadrotor to track the trajectory generated by the planning module. The detailed procedure of the planning module is as follows. Firstly, a safe position trajectory is obtained by an optimization-based method. The yaw safety corridor is designed to constrain the direction of the quadrotor utilizing the position trajectory. Next, a coarse-to-fine graph search strategy is proposed to search for a safe and perception-aware yaw angle path. Finally, by representing the trajectory as a uniform B-spline, a path-guided optimization method is put forward to quickly generate a safe, perception-aware and dynamically feasible trajectory. Above all, the yaw planning procedure is shown in Algorithm 1.

V. Coarse-to-Fine Path Search

Since cameras deployed on the UAVs usually have a limited field of view (FOV), the UAVs can not observe the omnidirectional obstacles. Therefore, a yaw safety corridor is proposed to ensure the safety of the yaw angle (see Section V-A). Then, a coarse-to-fine path search incorporating the yaw safety corridor is put forward to search for a safe and perception-aware yaw angle path (see Section V-B).

Algorithm 1 yaw Angle Planning

Input: The obstacle map $\mathcal{M}_o$, the active map $\mathcal{M}_a$, The start position point and the end position point p_s, p_e, The start yaw angle y_s.

Output: The position trajectory represented by B-spline $\boldsymbol{C}_t$ and the yaw angle trajectory $\boldsymbol{C}_y$;

$p_s, p_e, y_s \Leftarrow$ SetParam()
pos_traj $\Leftarrow$ PositionPlan($p_s, p_e, \mathcal{M}_o$)
if plan.type == YAW **then**
  searchNodeCor $\Leftarrow$ SamplePos(pos_traj)
  for all $node \in \mathcal{N}_1$ **do**
    $node$.info $\Leftarrow$ GetInfo($node$, $\mathcal{M}_a$)
  end for
  $yaw\_pathCor \Leftarrow$ Search($\mathcal{M}_o, searchNodeCor$)
  $searchNodeFine \Leftarrow$ SamplePos($yaw\_pathCor$)
  for all $node \in \mathcal{N}_2$ **do**
    $node$.info $\Leftarrow$ GetInfo($node$, $\mathcal{M}_a$)
  end for
  $yaw\_pathFine \Leftarrow$ Search($\mathcal{M}_o, searchNodeFine$)
  SetWaypoints($yaw\_pathFine$, y_s, y_e)
  $B\_spline \Leftarrow$ OptimizeTraj($cost1, dt$)
  $B\_spline \Leftarrow$ OptimizeTraj($cost2, dt$)
  Sample(B_spline, t)
else
  Sample($velocity\_dir, t$)
end if

A. Yaw Safety Corridor

Generally, the yaw angle is not refined during the path search and the trajectory optimization, which is a constant or along the tangent direction of the position trajectory. However, they do not fully exploit the environment information for the better localization. Therefore, it is essential to plan a perception-aware yaw angle trajectory.

The position trajectory is considered as safe when there is no intersection with the edge of the obstacle. However, there is no specific metric to evaluate the safety of a yaw angle trajectory. Therefore, a universal yaw safety metric is put forward to evaluate the safety of the yaw angle. Specifically, if the obstacles directly ahead are not in the view, the flight will be dangerous (see Fig. 5d). To this end, a yaw safety corridor is constructed, which includes all yaw angles that can see the front obstacles. As shown in Fig. 6, the blue part stands for the safe region and the view of the sensors can rotate in the blue region to keep safe. A yaw angle trajectory is considered as safe if it is totally limited in the yaw safety corridor.

B. Coarse-to-Fine Search

In the front-end, the position trajectory is uniformly sampled to obtain a series of positions $\boldsymbol{P} = \{\boldsymbol{p}_0, \boldsymbol{p}_1, \ldots \boldsymbol{p}_N\}$, in which $\boldsymbol{p}_l \in \mathbb{R}^3$ and $l \in [0, N]$. At each position except $\boldsymbol{p}_0$ and $\boldsymbol{p}_N$, K directions are sampled and the corresponding fisher information is calculated. Let $n_{i,j}$ denote the search node, which contains a sampled position and direction. Two near nodes from $n_{i,j}$ to $n_{i+1,s}$ are connected with a directed edge, where $i \in [0, N-1]$, $j, s \in [1, K]$. The weight of the edge

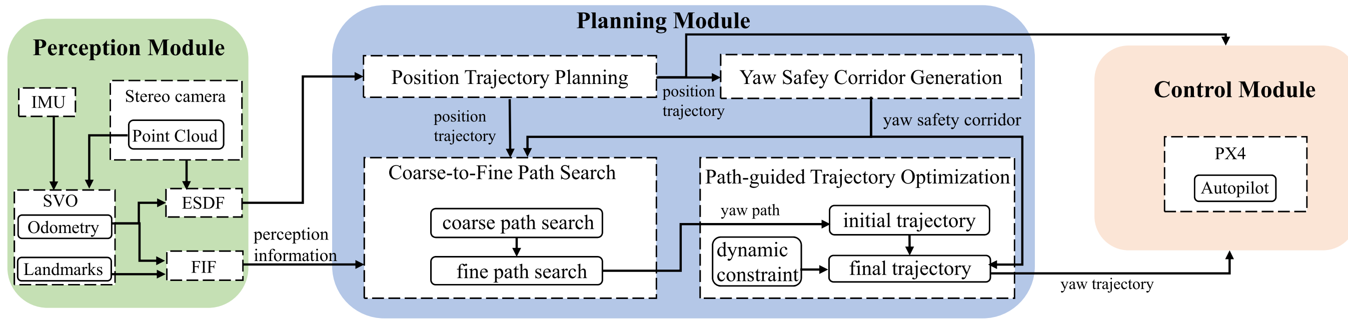


Fig. 4. System overview.

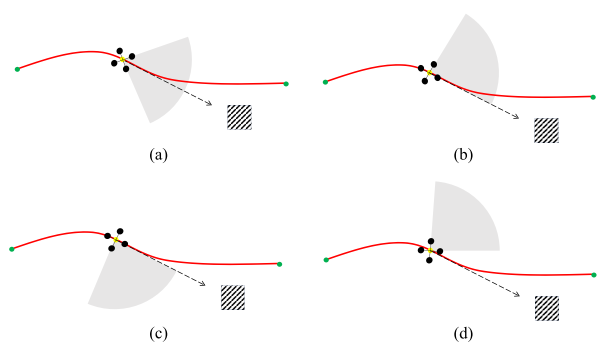


Fig. 5. The red line stands for the position trajectory. The gray sector represents the view of the UAV. The black line is the tangent line of the trajectory at this point. The textured grey cube is the obstacle directly in front of the UAV. (a): The view of the UAV is along the tangent direction of the position trajectory, and the UAV can see all the obstacles in the front. (b),(c): The bound of the sensor is exactly along the trajectory tangent line, and the UAV can see the obstacles directly ahead. (d): The UAV can not see the obstacles directly ahead at this rotation. It is a risky flight.

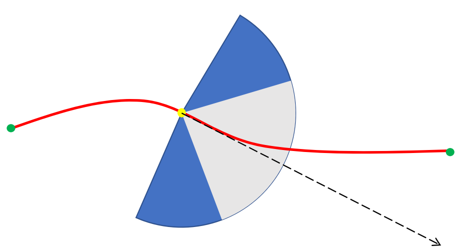


Fig. 6. Blue part stands for the safe region, the view of the sensors can rotate in the blue region to keep safe.

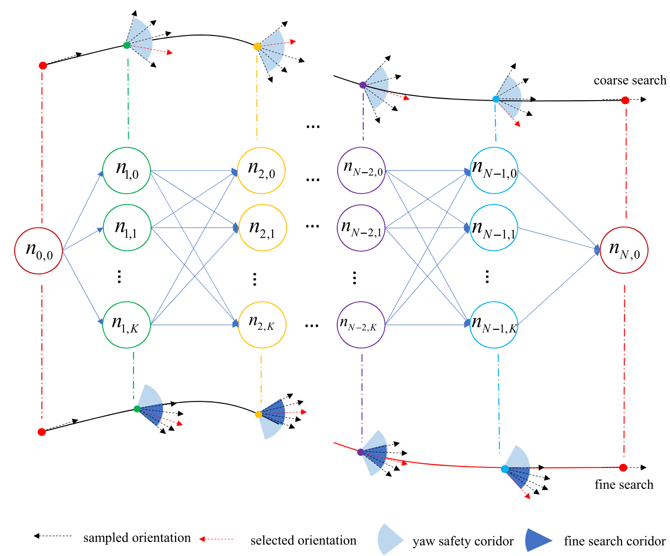


Fig. 7. The coarse-to-fine graph search process. $N+1$ points are sampled, including the start and the end points. At each point except the start and the end, $K+1$ directions in the yaw safety corridor are sampled at the coarse stage. Yaw angles are sampled in a relatively large region in the coarse graph search. Once the first path search is finished, a sequence of directions is obtained, which is represented by red arrows. Based on these yaw angles, $K+1$ directions, which are in the yaw safety corridor, are sampled at each point delicately. The second search will get a final path that is smooth, perception-aware and safe.

is set according to the smoothness and the fisher information. After that, a directed graph is obtained, which can be solved by $Dijkstra$ algorithm. To search a series of yaw angles which consider the safety, the smoothness and the localization accuracy, the objective cost function c_s is designed as

$$c_s = \sum_{i=1}^{N} \{\mu_s(\theta_{i,j} - \theta_{i+1,s})^2 + \mu_f c_f\}, \tag{4}$$

where $\mu_s \in \mathbb{R}$ is designed to penalize the change between two search nodes and $\theta_{i,j} \in \mathbb{R}$ is the jth sampling yaw angle at the ith position. $c_f \in \mathbb{R}$ is the cost of the fisher information and μ_f denotes the penalty coefficient. The fisher information threshold $v_{thr} \in \mathbb{R}$ is calculated by the method in [40]. We assume that the current pose localization is considered valid if at least M landmarks are present within the field of view of the camera. Initially, for a specific information metric, several sets of landmarks meeting the localization criteria are randomly sampled, and their average v_{thr} is employed as the threshold. Consequently, distinct thresholds can be associated when utilizing different information metrics. This approach ensures that the thresholds for various combinations of information metrics are automatically calculated and convey the same physical significance. Once the information exceeds a threshold, the improvement of the localization quality is limited. Therefore, the information cost objective function is as follows:

$$c_f(v_f) = \begin{cases} k_1 v_f + b, & v_f < 0, \\ k_2(v_f - v_{thr})^2, & 0 <= v_f < v_{thr}, \\ 0, & v_f >= v_{thr}, \end{cases} \tag{5}$$

where $v_f \in \mathbb{R}$ is the fisher information. $k_2 \in \mathbb{R}$ is chosen by the experiment as well as $k_1 \in \mathbb{R}$ and $b \in \mathbb{R}$ are chosen to keep the continuity.

Above all, the path search is formulated as a graph search problem. The sampling step of the fine path is generally small, which will result in many useless search nodes. Hence, we propose a coarse-to-fine search strategy. At the coarse stage, a sequence of nodes is roughly sought, and the sampling interval is relatively large. The objective function needs to promise the smoothness, which is designed as

$$c_{s_1} = \sum_{i=1}^{N} \{\mu_s(\theta_{i,j} - \theta_{i+1,s})^2 + \mu_f c_f\}. \tag{6}$$

As shown in Fig. 7, the sampling yaw angles of the fine stage are in the vicinity of directions obtained from the first stage. Since the sampling interval of the fine stage is small, we can ignore the smoothness. Then, a delicate path is devoted to improving the localization quality. The objective function at this stage is

$$c_{s_2} = \sum_{i=1}^{N} \mu_f c_f. \tag{7}$$

VI. Path-Guided Trajectory Optimization

A safe, smooth and perception-aware path is obtained after the front-end yaw angle path searching. However, it is hard to execute for UAVs. A more executable trajectory is obtained through a path-guided trajectory optimization method. During the optimization, an initial trajectory is firstly generated by the guidance of the yaw path. The first optimization is formulated as an unconstrained QP problem, which can directly obtain the closed-form solution. However, the trajectory is not exactly limited in the yaw safety corridor. In addition, it may be dynamically infeasible in some piecewise. Therefore, a second-step optimization is employed to obtain a better trajectory, wherein the gradient of the objective function pushes the trajectory to the safe area. The solution of the first stage optimization is the initial value of the second stage optimization. A p_d degree uniform B-spline is used to represent the trajectory. The control points are defined as $\{\ Q_0, Q_1, \ldots, Q_N\ \}$. The B-spline is contained within the convex hull of the control polyline, which can be employed to limit the B-spline within the yaw safety corridor. Besides, the piecewise B-spline can keep the derivative continuity. A two-step path-guided trajectory optimization method is proposed to efficiently refine the trajectory. During the optimization, an initial trajectory is firstly generated by the guidance of the yaw path, which considers the smoothness. The initial trajectory can not keep the front obstacles in the view. Therefore, the initial trajectory is guided to the yaw safety corridor in the second phase. The smoothness objective function is designed as

$$f_s = \sum_{i=1}^{N-1} \|(Q_{i+1} - Q_i) + (Q_{i-1} - Q_i)\|^2, \tag{8}$$

$f_s \in \mathbb{R}$ is designed as an elastic bond cost function. The shape of the B-spline is controlled by its control points. Hence, it is a straight line if f_s is zero, which is ideally smooth. A series of yaw angles are obtained after the front-end search, which are utilized to generate a guided B-spline. Each control point Q_i is assigned with a corresponding control point $L_i \in \mathbb{R}$ that is uniformly sampled along the guided B-spline. Then $f_g \in \mathbb{R}$ is defined as the sum of the Euclidean distance between these point pairs, which can be expressed as

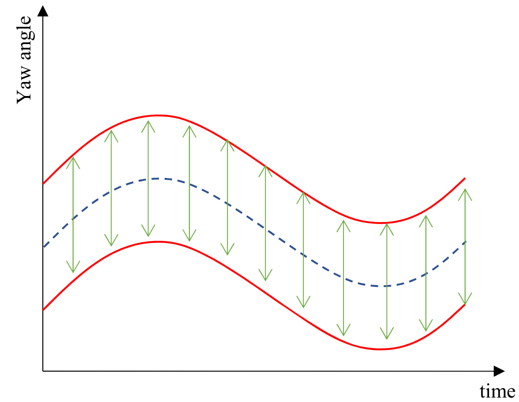


Fig. 8. The horizontal axis represents time, and the longitudinal axis represents yaw angles. The blue line is calculated by the position trajectory along the tangent direction. Red lines are the upper and down bounds of the planning trajectory. The final trajectory, which is in the middle of two lines, is considered safe.

$$f_g = \sum_{i=0}^{N} \|Q_i - L_i\|^2. \tag{9}$$

Although a safe yaw path is obtained in the search stage, the back-end optimization will make the B-spline unsafe. Therefore, a cost function $f_c \in \mathbb{R}$ is designed to penalize the points beyond the yaw safety corridor to make the trajectory safe (see Fig. 8), which can be expressed as

$$f_c = \sum_{i=1}^{N} c_c(Q_i), \tag{10}$$

$$c_c(Q_i, Q_{ni}) = \begin{cases} |Q_i - Q_{ilower}|^2, & Q_i \le Q_{ilower}, \\ |Q_i - Q_{iupper}|^2, & Q_i \ge Q_{iupper}, \\ 0, & else, \end{cases} \tag{11}$$

where $Q_{ilower} \in \mathbb{R}$ and $Q_{iupper} \in \mathbb{R}$ are the lower limit and the upper limit of the yaw safety corridor, respectively. Besides, a cost function similar to (11) is adopted to penalize the velocity and the acceleration which exceed the maximum given value

$$c_v(v_i) = \begin{cases} (V_i^2 - {V_{max}}^2)^2, & |V_i| \ge V_{max}, \\ 0, & |V_i| < V_{max}. \end{cases} \tag{12}$$

$$c_a(A_i) = \begin{cases} (A_i^2 - {A_{max}}^2)^2, & |A_i| \ge A_{max}, \\ 0, & |A_i| < A_{max}, \end{cases} \tag{13}$$

where V_i and A_i are the velocity and the acceleration control points being calculated as

$$V_i = \frac{1}{\Delta t}(Q_{i+1} - Q_i), \tag{14}$$

$$A_i = \frac{1}{\Delta t}(V_{i+1} - V_i). \tag{15}$$

Utilizing the convex hull property, $f_v \in \mathbb{R}$ and $f_a \in \mathbb{R}$ are defined to penalize the infeasible the velocity and the acceleration control points

$$f_v = \sum_{i=0}^{N-1} c_v(V_i), \tag{16}$$

$$f_a = \sum_{i=0}^{N-2} c_a(A_i), \tag{17}$$

Above all, the overall objective function is built as

$$f = \lambda_s f_s + \lambda_l f_l + \lambda_c f_c + \lambda_v f_v + \lambda_a f_a. \tag{18}$$

A two-step optimization strategy is proposed to efficiently refine the trajectory. In the first step, we employ a guided path to attract the trajectory and ensure the smoothness. In this stage, the fisher information is fully used to localize more accurately. The objective function is

$$f_1 = \lambda_{s1} f_s + \lambda_{g1} f_g, \tag{19}$$

where $\lambda_{s1} \in \mathbb{R}$ is the smoothness coefficient and $\lambda_{g1} \in \mathbb{R}$ denotes the cost coefficient to penalize the difference between the guided B-spline trajectory and the final B-spline trajectory. Accordingly, we can obtain a smooth and perception-aware trajectory. However, the trajectory is not exactly limited in the yaw safety corridor. In addition, it may be dynamically infeasible in some piecewise. Therefore, a second-step optimization is employed to get a better trajectory, wherein the gradient of the objective function pushes the trajectory to the safe area. In this stage, the objective function is designed as

$$f_2 = \lambda_{s2} f_s + \lambda_{l2} f_l + \lambda_{v2} f_c + \lambda_{a2} f_a, \tag{20}$$

which combines the punishment for the roughness, unsafety and dynamically infeasibility. After that, we can obtain a safe, smooth, perception-aware and dynamically feasible trajectory. It can effectively generate trajectories with low localization uncertainty when employing Fisher Information Field directly for gradient-based optimization. However, it also presents the challenge of potential entrapment in local minima, which can result in suboptimal solutions. Path-guided methods offer a solution to solve the problem of getting stuck in local minima. Besides, path-guided method leverages a safe path found by the front-end search, thus obtaining a safe trajectory. However, unsafe trajectories may be generated when employing Fisher Information Field directly for gradient-based optimization.

Remark 1: Although the proposed method involves an extra optimization step, it can generate more efficient trajectories in a shorter time. On the one hand, the initial phase of optimization is formulated as an unconstrained quadratic problem, which can directly obtain the closed-form solution. As a result, the computation time is more efficient. On the other hand, based on the initial trajectory, fewer convergence times are required in the second stage compared with optimizing the trajectory directly.

VII. Results

We conduct both the simulation and real-world experiments to prove the effectiveness of the proposed yaw safety corridor. In addition, perception awareness, failure rate and localization accuracy are evaluated in three methods. The simulation visibility graph of the Fisher Information Field is shown in Fig. 9. Colored dots represent landmarks, with different colors indicating different heights of the landmarks. Colored arrows represent the optimal orientation under the current voxel, with different colors representing different fisher information. The distribution of textures in the environment can be observed from Fig. 9.

A. Yaw Safety Corridor

We compare the proposed SLUP strategy with the FIF-based perception-aware planning method (FIFP) presented in [40] and a baseline method that does not consider perception-awareness (NP) [42]. Once the position trajectory is generated, the yaw safety corridor is determined. As shown in Fig. 10, four groups of simulations are conducted under the different scenes. As defined above, the yaw safety corridor is generated according to the tangent direction of the sampled points. Therefore, the NP planning totally satisfies the requirement. The FIFP utilizes the trajectory optimization method to generate a trajectory that completely takes advantage of the fisher information. However, it does not consider the safety, which may lead to a fatal disaster. In the FIFP, the views of UAVs are not always in the yaw safety corridor. Even at some points, the view is against the tangent direction of the corresponding points, which is impractical during the flight.

In our method, in the front-end search process, the yaw safety corridor is seen as a hard constraint to keep the yaw path exactly satisfying the safety constraint, while in the back-end, safety is seen as a soft constraint to push the trajectory into the yaw safety corridor. The same with obstacle avoidance, it can not always satisfy the safety constraint during the trajectory optimization. However, when the span of the B-spline is small, the whole trajectory can be seen as safe if the control points satisfy the safety constraint. The span between two control points are tunable parameters decided by the parameterization of the B-spline. The trajectory may be unsafe in some extreme cases such as the environment with dense obstacles. In this situation, a smaller span can be employed to re-parameterize the B-spline to meet the safety constraint. Although the reduction in span results in an increase in the number of control points, the optimization time also increases. However, in the experiments, with the use of the adjustment strategy, the optimization time does not exceed 10ms, thus ensuring the real-time performance. At the same time, in an optimization problem, the coefficients of the optimization terms also influence the results. For different planning scenes, we try our best to choose the most balanced coefficients. As shown in Fig. 11, the yaw angle trajectories under four scenes are drawn. The upper limit and the lower limit of the yaw safety corridor are calculated by the NP and the FOV. It is clear that the FIFP can not limit the yaw angle in the yaw safety corridor. However, our method ideally

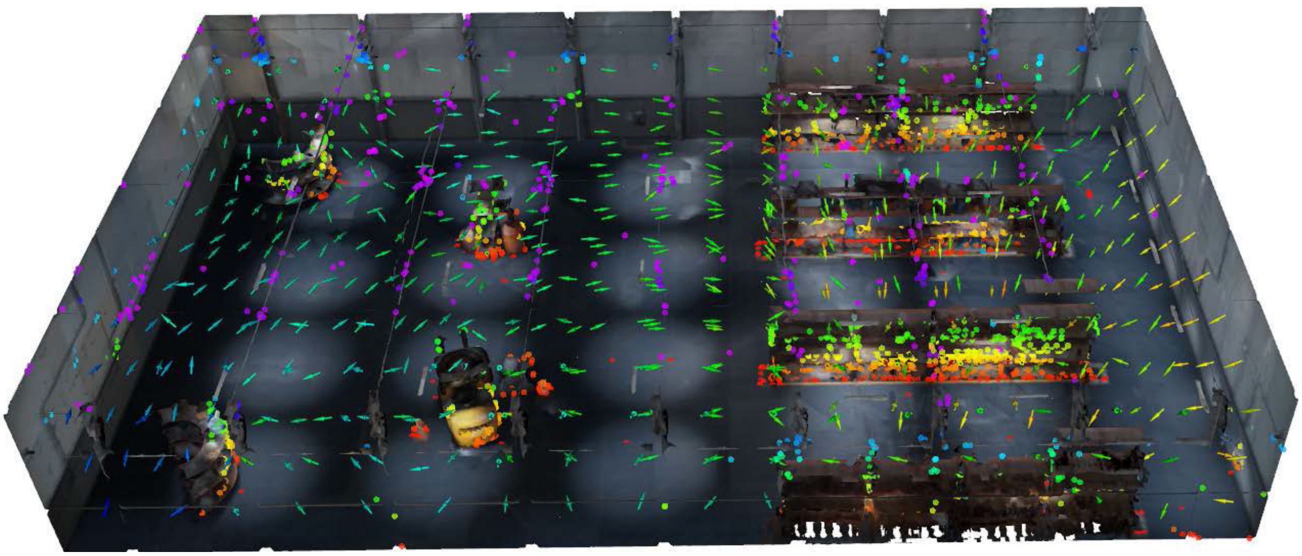


Fig. 9. The simulation visibility graph of the Fisher Information Field.

limits the yaw angle in the yaw safety corridor. Besides, in the simulation, a coarse-to-fine search approach processes an order of magnitude fewer nodes compared to the regular search method, resulting in a faster search.

B. Perception Awareness and Failure Rate

Intuitively, the number of the visual features reflects the amount of perception information obtained by the quadrotor, which is represented by colored dots in Fig. 9. More theoretically, the fisher information is used for evaluating the localization uncertainty. Therefore, the Fisher Information Field is constructed, which is visualized by the colored arrows. Specifically, the direction of the arrow represents the best sampling view in this voxel and the deeper the color, the richer the perception information. By observing the sampling direction and the visibility of the FIF, the perception awareness of the trajectory can be analyzed qualitatively. The quantitative results under four scenes are listed in Tab. II. Poses are uniformly sampled along a trajectory and the average fisher information cost is calculated according to (5). A sampled yaw angle will be seen as a failure if it does not satisfy the safety constraint. A comprehensive indicator that takes into account perception awareness and safety (CIPS) is defined to measure the performance:

$$CIPS = \lambda_{info} c_f + \lambda_{fail} r_u, \tag{21}$$

where $\lambda_{info} \in \mathbb{R}$ and $\lambda_{fail} \in \mathbb{R}$ represent the weight of the fisher information cost and the failure rate, respectively. Moreover, c_f represents the fisher information cost and r_u represents the failure rate. The trajectory generated by the FIFP method will produce the lowest fisher information cost. The trajectory generated by the NP will make zero failure rate because of the definition of the yaw safety corridor. Meanwhile, the FIFP method has the best perception awareness performance, and the NP performs worst. However, the proposed method perfectly balances the perception quality and the safety, which provides the best comprehensive performance.

TABLE II
COMPARISON OF DIFFERENT INDICATORS

Scene	Method	Info Cost	Failure	CIPS
scene 1	FIFP	**70.93**	51.52	61.23
	NP	84.95	**0**	42.46
	SLUP	80.67	**0**	**40.34**
scene 2	FIFP	**71.29**	41.41	56.35
	NP	87.1	**0**	43.55
	SLUP	77.82	**0**	**38.91**
scene 3	FIFP	**69.76**	78.79	74.28
	NP	84.92	**0**	42.46
	SLUP	82.07	**0**	**41.04**
scene 4	FIFP	**55.67**	75.76	65.72
	NP	91.64	**0**	45.82
	SLUP	81.8	**0**	**40.9**

TABLE III
LOCALIZATION ERROR

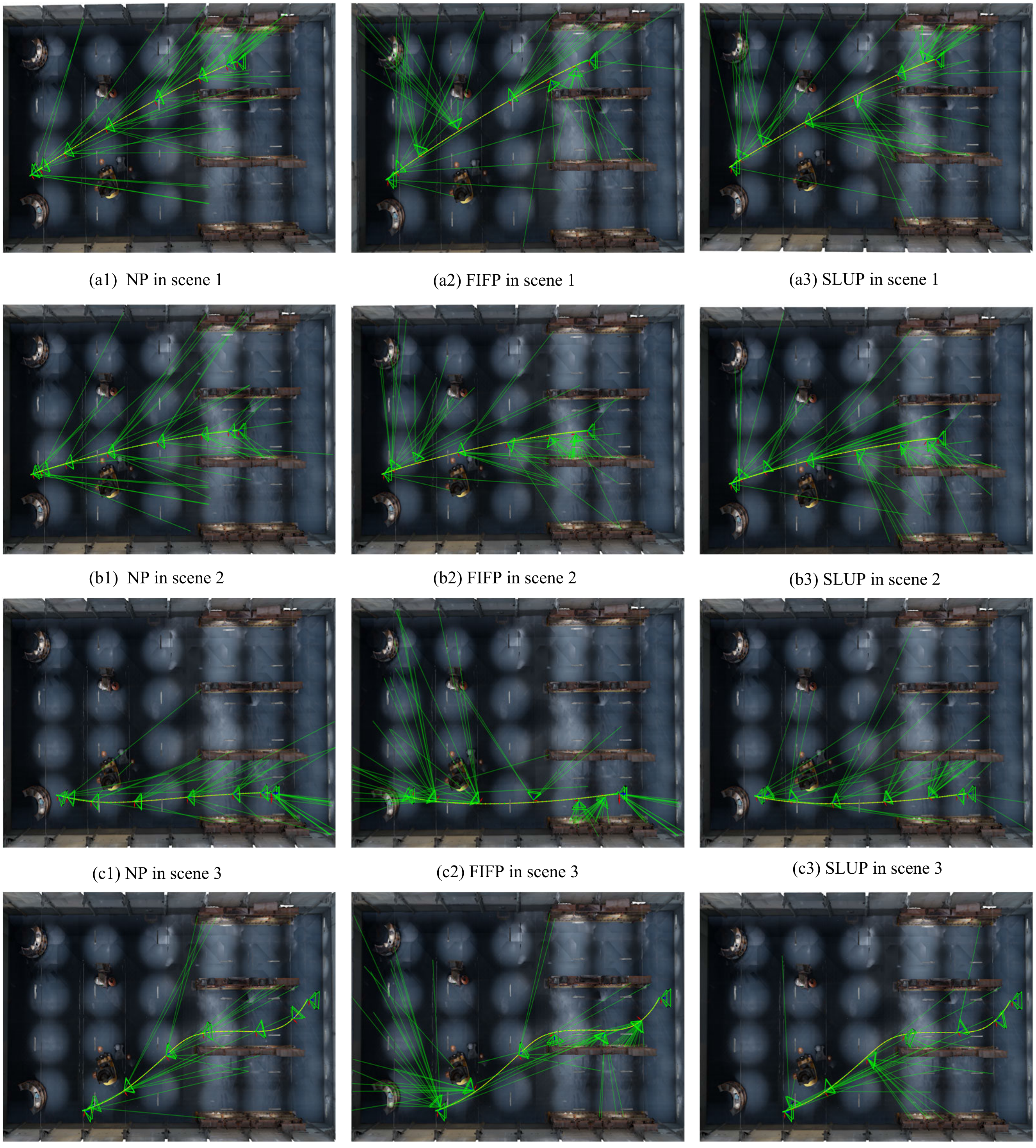
Method	position error[m]	angle error[deg]
FIFP	**0.25**	**1.18**
NP	0.4	1.43
SLUP	0.29	1.26

C. Localization Accuracy

After getting a trajectory, some poses are sampled, and the corresponding images are rendered at these poses to evaluate the localization accuracy. The average localization error results of the three methods are calculated in scene 4, which are listed in Tab. III. As expected, the NP method performs worst while the FIFP method has the most accurate localization due to the abundant fisher information. The localization quality of the proposed method is close to the FIFP, which considers the perception awareness and the safety simultaneously.

D. Real-World Experiments

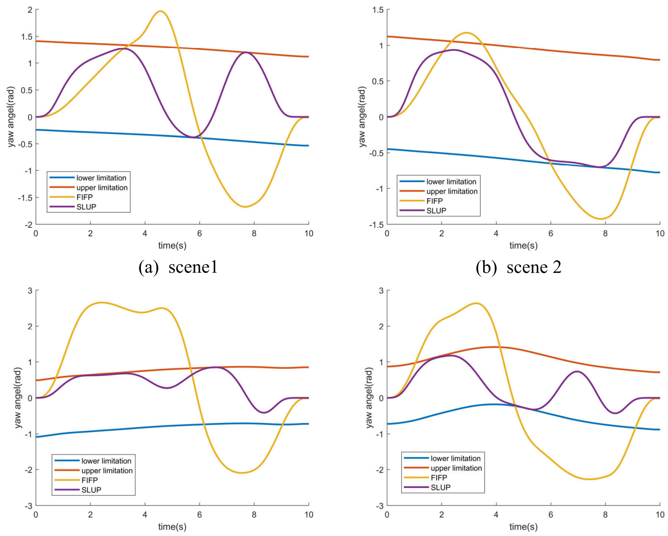
In this section, a self-made quadrotor is employed to evaluate the performance of the proposed approach in an



(a1) NP in scene 1
(a2) FIFP in scene 1
(a3) SLUP in scene 1
(b1) NP in scene 2
(b2) FIFP in scene 2
(b3) SLUP in scene 2
(c1) NP in scene 3
(c2) FIFP in scene 3
(c3) SLUP in scene 3
(d1) NP in scene 4
(d2) FIFP in scene 4
(d3) SLUP in scene 4

Fig. 10. Simulation results under four scenes. In (a1), (b1), (c1), (d1), the yaw angles are generated by the NP. In (a2), (b2), (c2), d2), the yaw angles are generated by the FIFP. In (a3), (b3), (c3), (d3), the yaw angles are generated by our method SLUP. In the NP, the yaw angle is always along the tangent direction of the position trajectory, resulting in the big localization uncertainty. In the FIFP, the safety is not considered, and the MAV often can not see the obstacles directly ahead. In (c2), a view totally opposite to the tangent direction of the trajectory is sampled. It is impractical in the actual flight. In our method, yaw angles can be bound in the yaw safety corridor.

indoor environment. As shown in Fig. 12, the experimental site is with a size of 10m × 6m × 2.5m. In addition, the quadrotor is equipped with an intel microcomputer (Intel Nuc with the Core i7-10710U) to run the proposed method. An intel realsense stereo camera D455 equipped on the head of the quadrotor is utilized to build ESDF and provide point clouds information for SVO [13]. SVO algorithm can generate landmarks to build FIF and offer the odometry for



(a) scene1 (b) scene 2 (c) scene 3 (d) scene 4

Fig. 11. The variation of yaw angle with time under four scenes. The red line represents the upper limit of the yaw safety corridor, and the blue line represents the lower limit. The purple line represents the trajectory generated by the proposed method. The orange line represents the yaw corridor generated by the FIFP.

Fig. 12. The quadrotor platform and experiment field.

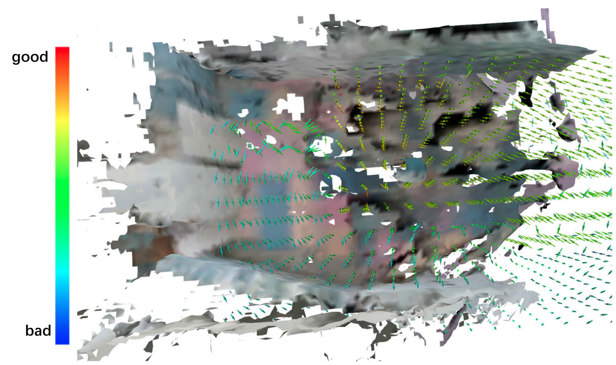


Fig. 13. The visibility of FIF. 200 orientations are sampled at each voxel grid, the optimal orientation is chosen according to fisher information. The optimal orientations are represented by the colorful arrows and the color indicates the localization quality.

the localization. An open-source PIXHAWK4 autopilot is equipped to receive the control commands.

We first build ESDF for the obstacles avoidance and FIF for perception-aware planning. The visibility of FIF is shown in Fig. 13. The voxel size of ESDF and FIF is set as 0.05m to promise the mapping quality. The planning trajectory is executed after the quadrotor hovers at 1m. The maximum speed is set to 0.5 m/s during the flight. Other parameters are the same as in the simulation environment. As shown in

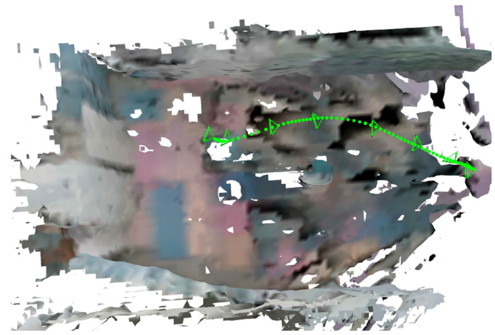
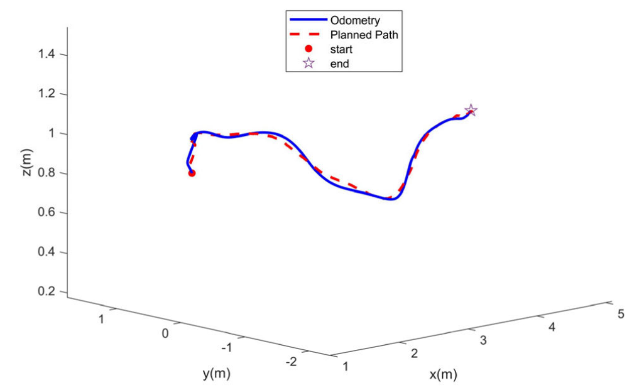
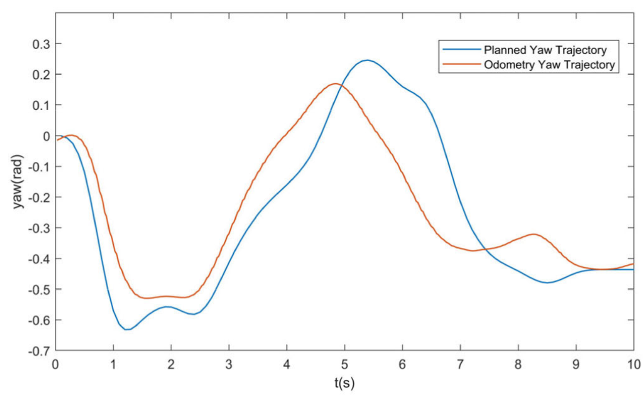


Fig. 14. The real-world experiment, the visible field is built by voxblox and the trajectory is planned by the proposed method. The green line is the position trajectory and the green cone represents the yaw angle direction.



(a)



(b)

Fig. 15. The position and yaw trajectory of the odometry and the expectation.

Fig. 14, the green line indicates the sampled position trajectory, and the green sector cone the yaw trajectory sampled at 8 points. The planned and the actual trajectory of the quadrotor are depicted in Fig. 15. More experimental details can be found in the attached video.

VIII. Conclusion

This article studies the factors related to the yaw angle safety and perception quality in the motion planning for the quadrotors. A safety-driven and localization uncertainty-driven method is proposed to avoid obstacles directly along the tangent direction of the trajectory. Besides, it can also actively obtain more information in the environment to improve the localization accuracy. A coarse-to-fine strategy is designed to search for a safe and perception-aware yaw angle path. After that, the convex hull property is utilized to generate a safe and perception-aware trajectory by formulating the trajectory as a uniform B-spline. Finally, comparative simulation and real-world experiments are carried out to verify the superior performance of the proposed method. In our future work, we will extend our safe and perception-aware planning method to autonomous swarms.

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